

Birhane Hiwot Sunday School — QA Test Case Document

Application: Birhane Hiwot Sunday School Attendance Management System

Version: 0.1.0

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Related assets: [docs/qa/Birhane-Hiwot-API.postman_collection.json](#), [docs/qa/Birhane-Hiwot.postman_environment.json](#)

1. Test Scope

In scope	Out of scope
Authentication & session management	Third-party Google Sheets sync reliability (unless configured)
Role-based page access (middleware)	Load/performance benchmarking
Student CRUD & registration workflow	Mobile native apps
Attendance marking & reporting	Penetration testing
Facilitator & admin user management	
Teacher, subject & assignment management	
Student results & grading	
Payment status tracking	
Student request approval flow	
QR code generation & verification	
REST API endpoints	

2. Test Environment

Item	Value
Base URL (local)	http://localhost:3000
Base URL (staging)	<i>Set before test run</i>
Database	MongoDB
Auth	NextAuth (Credentials, JWT session, 24h expiry)
Calendar	Ethiopian calendar for dates & academic year

Test Accounts Required

Role	Purpose
Super Admin	Full system access
HR Admin	HR module, attendance facilitators, limited admin routes
Education Admin	Education module, teachers, subjects
Attendance Facilitator	Mark attendance, optional student creation
Education Facilitator	Results & education reports
Teacher	Subject/grade assignment (if UI enabled)

3. Test Case Summary

Module	Total Cases	Priority High
Authentication	12	10
Authorization / RBAC	18	16
Students	28	20
Attendance	16	14
Facilitators	14	10
Teachers & Subjects	16	10
Results	12	8
Payments	8	6
Student Requests	10	8
QR Code	6	5
Reports & Export	8	4
Total	148	111

4. Authentication & Session

ID	Test Case	Steps	Expected Result	Priority
AUTH-001	Valid login	Enter valid email/password on <code>/login</code> , submit	Redirect to role dashboard; session cookie set	High
AUTH-002	Invalid password	Enter valid email, wrong password	Error message; remain on login page	High
AUTH-003	Invalid email	Enter non-existent email	Error message; no session created	High

ID	Test Case	Steps	Expected Result	Priority
AUTH-004	Empty fields	Submit with blank email or password	Validation prevents submit or shows error	High
AUTH-005	Case-insensitive email	Login with different email casing than stored	Login succeeds	Medium
AUTH-006	Session persistence	Login, refresh page	User remains authenticated	High
AUTH-007	Session expiry	Wait 24h or invalidate token	User redirected to login	Medium
AUTH-008	Logout	Click logout	Session cleared; redirected to login	High
AUTH-009	Protected route without auth	Visit <code>/admin/dashboard</code> without login	Redirect to <code>/login</code>	High
AUTH-010	API session check	GET <code>/api/auth/session</code> after login	Returns user id, email, role	Medium
AUTH-011	CSRF token flow	GET <code>/api/auth/csrf</code> then login via credentials callback	Login succeeds with valid CSRF	Medium
AUTH-012	Super Admin setup (one-time)	POST <code>/api/setup/super-admin</code> with <code>confirm: CREATE_SUPER_ADMIN</code>	Creates Super Admin if none exists	High

5. Authorization (Role-Based Access)

ID	Test Case	Role	Route / Action	Expected Result	Priority
RBAC-001	Super Admin full access	Super Admin	<code>/super-admin/*</code>	Access granted	High
RBAC-002	Non-super blocked from super-admin	HR Admin	<code>/super-admin/dashboard</code>	Access denied / redirect	High
RBAC-003	HR Admin HR module	HR Admin	<code>/hr/*</code>	Access granted	High
RBAC-004	HR Admin blocked from education	HR Admin	<code>/education/*</code>	Access denied	High

ID	Test Case	Role	Route / Action	Expected Result	Priority
RBAC-005	Education Admin education module	Education Admin	/education/*	Access granted	High
RBAC-006	Attendance Facilitator attendance	Attendance Facilitator	/facilitator/attendance/*	Access granted	High
RBAC-007	Attendance Facilitator results blocked	Attendance Facilitator	/facilitator/results/*	Access denied	High
RBAC-008	Education Facilitator results	Education Facilitator	/facilitator/results/*	Access granted	High
RBAC-009	HR Admin limited admin routes	HR Admin	/admin/students, /admin/facilitators, /admin/reports	Access granted	High
RBAC-010	HR Admin blocked from other admin	HR Admin	/admin/dashboard (if restricted)	Access denied	Medium
RBAC-011	Education Admin limited admin	Education Admin	/admin/facilitators, /admin/reports	Access granted	High
RBAC-012	Facilitators API — list	Super Admin / HR Admin	GET /api/facilitators	200 with facilitator list	High
RBAC-013	Facilitators API — forbidden	Unauthenticated / wrong role	GET /api/facilitators	403 Forbidden	High
RBAC-014	Teachers API — manage	Education Admin	POST /api/teachers	201 Created	High
RBAC-015	Teachers API — forbidden	Attendance Facilitator	POST /api/teachers	403 Forbidden	High
RBAC-016	Admin users — Super	HR Admin	GET /api/admin-users	403 Forbidden	High

ID	Test Case	Role	Route / Action	Expected Result	Priority
Admin only					
RBAC-017	Admin users — Super Admin only	Super Admin	GET <code>/api/admin-users</code>	200 OK	High
RBAC-018	Permission refresh without re-login	HR toggles <code>canAddStudent</code>	Facilitator session still active	Updated permission reflected in session	Medium

6. Student Management

ID	Test Case	Steps	Expected Result	Priority
STU-001	List all students	GET <code>/api/students</code>	Returns array of students (no photo/qr in list)	High
STU-002	Filter by grade	GET <code>/api/students?grade=ᠠᠨᠠᠭᠤ ᠰᠤᠶ᠋ᠠᠨ</code>	Only matching grade returned	High
STU-003	Filter by academic year	GET <code>/api/students?academicYear=2018</code>	Filtered results	High
STU-004	Get by Unique ID	GET <code>/api/students?uniqueId=BH-001</code>	Single student or 404	High
STU-005	Get by Mongo ID	GET <code>/api/students/{id}</code>	Full student record	High
STU-006	Create student — Super Admin	POST with required fields + <code>userRole: Super Admin</code>	201, <code>_id</code> returned, QR generated if configured	High
STU-007	Create student — missing required field	Omit <code>Unique_ID</code>	400 with field name in error	High
STU-008	Create duplicate Unique_ID	POST same <code>Unique_ID</code> twice	Second request handled (lock prevents duplicate)	High
STU-009	Facilitator without permission	POST as Attendance Facilitator, <code>canAddStudent: false</code>	403 <code>ADD_STUDENT_DENIED</code>	High
STU-010	Facilitator with permission	HR enables <code>canAddStudent</code> , facilitator creates student	201 Created	High
STU-011	Facilitator wrong grade	Create student for unassigned grade	403 <code>RESTRICTED_GRADE</code>	High

ID	Test Case	Steps	Expected Result	Priority
STU-012	Invalid photo format	POST with non-image <code>photo_data_url</code>	400 validation error	Medium
STU-013	Update student (PUT)	PUT <code>/api/students/{id}</code> with changed phone	200, fields updated	High
STU-014	Regenerate QR on Unique_ID change	Change <code>Unique_ID</code> via PUT	New QR code generated	Medium
STU-015	Update photo (PATCH)	PATCH with valid JPEG data URL	Photo updated	Medium
STU-016	Generate QR (PATCH)	PATCH with <code>generateQR: true</code>	QR data URL in response	Medium
STU-017	Check duplicate — exists	POST <code>/api/students/check-duplicate</code> with matching identity	<code>{ exists: true }</code>	High
STU-018	Check duplicate — not exists	POST with new identity	<code>{ exists: false }</code>	High
STU-019	Check duplicate — missing fields	Omit <code>Sex</code>	400 error	Medium
STU-020	Delete student	DELETE <code>/api/students</code> with valid <code>id</code>	200, student removed	High
STU-021	Delete invalid ID	DELETE with bad ObjectId	400 error	Medium
STU-022	Student form — Regular classification	UI: select Regular, fill all sections	Grade options show standard grades	High
STU-023	Student form — Extension classification	Select Extension	Shows Level 1 / Level 2 grades	Medium
STU-024	Student form — Sign Language	Select Sign Language	Shows sign language grade option	Medium
STU-025	Student form — Summer	Select Summer	Shows Grade 7 morning/afternoon options	Medium
STU-026	Student details view	Open student profile page	All personal, academic, attendance tabs visible	High
STU-027	Student count by grades	POST <code>/api/students/count</code>	Correct count for grade + year	Medium

ID	Test Case	Steps	Expected Result	Priority
STU-028	Total students endpoint	GET <code>/api/students/total</code>	Returns total count	Low

7. Attendance

ID	Test Case	Steps	Expected Result	Priority
ATT-001	List attendance by date	GET <code>/api/attendance?date=YYYY-MM-DD</code>	Attendance records for date	High
ATT-002	Filter by grade	GET with <code>grade + academicYear</code>	Only students in that grade	High
ATT-003	Attendance summary	GET with <code>summary=true</code>	<code>{ total, present }</code> counts	High
ATT-004	Mark present	POST bulk attendance with <code>present: true</code>	200, upsert success	High
ATT-005	Mark absent with permission	POST with <code>present: false, hasPermission: true, reason</code>	Saved correctly	High
ATT-006	Mark absent without permission	POST with <code>present: false, hasPermission: false</code>	Saved correctly	High
ATT-007	Idempotent re-submit	POST same date/student twice	Updates existing row, no duplicate	High
ATT-008	Invalid payload	POST without <code>date</code> or <code>attendance</code> array	400 error	High
ATT-009	Student attendance history	GET <code>/api/attendance/{studentId}</code>	History for student	Medium
ATT-010	UI — mark all present	Use "Mark All Present" in attendance UI	All students marked present	High
ATT-011	UI — QR scan attendance	Scan valid student QR	Student marked present	High
ATT-012	UI — invalid QR	Scan tampered/invalid QR	Error shown, no attendance marked	High
ATT-013	UI — date picker (Ethiopian)	Select Ethiopian date	Correct Gregorian mapping used	Medium

ID	Test Case	Steps	Expected Result	Priority
ATT-014	UI — Sundays only mode	Set ATTENDANCE_CALENDAR_MODE to sundays_only	Non-Sunday dates disabled	Medium
ATT-015	Temp attendance endpoint	GET /api/attendance/temp	Returns draft/temp data	Low
ATT-016	Export attendance report	Generate Excel/PDF from reports page	File downloads with correct data	Medium

8. Facilitators

ID	Test Case	Steps	Expected Result	Priority
FAC-001	List facilitators	GET /api/facilitators (authenticated, authorized)	List without passwords	High
FAC-002	Get facilitator by ID	GET /api/facilitators?id={id}	Single record	High
FAC-003	Create Attendance Facilitator	POST with name, email, password, role, grade	201 Created	High
FAC-004	Create without grade	POST Attendance Facilitator without grade	400 "Grade is required"	High
FAC-005	Duplicate email	POST with existing email	409 Conflict	High
FAC-006	Update facilitator	PUT with changed name/grade	200 Updated	High
FAC-007	Delete facilitator	DELETE with valid id	200 Deleted	High
FAC-008	HR Admin creates attendance facilitator only	HR Admin POST Education Facilitator	403 No permission	High
FAC-009	Toggle canAddStudent	PATCH /api/facilitators/{id} {canAddStudent: true }	Field updated	High
FAC-010	Education facilitator CRUD	Use /api/education-facilitators as Education Admin	CRUD succeeds	High
FAC-011	Education facilitator — forbidden	Attendance Facilitator calls education-facilitators API	403 Forbidden	Medium
FAC-012	UI — add facilitator form	Fill add facilitator page	Facilitator appears in list	High

ID	Test Case	Steps	Expected Result	Priority
FAC-013	UI — edit facilitator	Edit name/email/grade	Changes saved	High
FAC-014	Total facilitators count	GET <code>/api/facilitators/total</code>	Returns count	Low

9. Teachers, Subjects & Assignments

ID	Test Case	Steps	Expected Result	Priority
TCH-001	List teachers	GET <code>/api/teachers</code> as Education Admin	Teacher list returned	High
TCH-002	Create teacher	POST with email (password optional, defaults)	201 Created	High
TCH-003	Update teacher	PUT with new assignedSubjects	200 Updated	Medium
TCH-004	Delete teacher	DELETE with valid id	200 Deleted	High
TCH-005	List subjects	GET <code>/api/subjects?academicYear&grade</code>	Subject groups returned	High
TCH-006	Add subject to group	POST <code>/api/subjects</code> with name, grade, year	201, subject added to group	High
TCH-007	Duplicate subject	POST same subject for same grade/year	409 Conflict	Medium
TCH-008	Batch add subjects	POST <code>/api/subjects/batch</code>	Multiple subjects created	Medium
TCH-009	Delete subject group	DELETE <code>/api/subjects/{id}</code>	Group removed	Medium
TCH-010	List teacher assignments	GET <code>/api/teacher-assignments</code>	Assignments list	Medium
TCH-011	Create assignment	POST with teacherId, subjectId, grade, year	201 Created	Medium
TCH-012	Duplicate assignment	POST same teacher/subject/grade/year	409 Conflict	Medium
TCH-013	Delete assignment	DELETE <code>/api/teacher-assignments/{id}</code>	Assignment removed	Medium

ID	Test Case	Steps	Expected Result	Priority
TCH-014	UI — manage teachers page	Add/edit teacher in education module	Changes reflected	High
TCH-015	UI — subjects page	Add subjects for grade/year	Subjects appear in list	High
TCH-016	UI — teacher attendance	Mark teacher attendance in education module	Record saved	Medium

10. Student Results

ID	Test Case	Steps	Expected Result	Priority
RES-001	List all results	GET <code>/api/student-results</code>	Results array	High
RES-002	Filter by student	GET <code>?studentId={id}</code>	Student's results only	High
RES-003	Create result	POST with scores (A1, A2, mid, final)	201, total & letter grade calculated	High
RES-004	Duplicate result	POST same student/subject/year	409 Conflict	High
RES-005	Grade calculation A+	Total score ≥ 90	Grade = "A+"	Medium
RES-006	Grade calculation F	Total score < 45	Grade = "F"	Medium
RES-007	Update result	PUT <code>/api/student-results/{id}</code>	Scores & grade recalculated	High
RES-008	Delete result	DELETE <code>/api/student-results/{id}</code>	Result removed	Medium
RES-009	UI — enter results	Fill results form in education/facilitator module	Result saved & displayed	High
RES-010	UI — results tab on student	Open student profile → Results tab	Results listed per subject	High
RES-011	UI — export results report	Export from reports page	PDF/Excel generated	Medium
RES-012	Missing required fields	POST without studentId	400 error	High

11. Payments

ID	Test Case	Steps	Expected Result	Priority
PAY-001	Get payment status — new student	GET <code>/api/payment?year&studentId</code>	All Ethiopian months initialized "Not Paid"	High
PAY-002	Get payment — missing params	GET without year or studentId	400 error	High
PAY-003	Save payment status	POST with month data (Paid + amount)	200 Saved	High
PAY-004	Partial month data	POST with only some months	Other months normalized to Not Paid	Medium
PAY-005	UI — payment tab	Open student → Payment Status tab	Monthly grid displayed	High
PAY-006	UI — mark month paid	Toggle month to Paid, enter amount, save	Status persisted on reload	High
PAY-007	UI — mark month not paid	Change Paid back to Not Paid	Status updated	Medium
PAY-008	Get student payments route	GET <code>/api/students/{id}/payments</code>	Payment history returned	Medium

12. Student Requests

ID	Test Case	Steps	Expected Result	Priority
REQ-001	Submit request	POST <code>/api/student-requests</code> with studentData	201, status = pending	High
REQ-002	List pending requests	GET <code>?status=pending</code>	Pending requests only	High
REQ-003	Get request by ID	GET <code>/api/student-requests/{id}</code>	Full request detail	High
REQ-004	Approve request	PATCH with status=approved	Status updated	High
REQ-005	Reject request	PATCH with status=rejected + reason	Status & reason saved	High
REQ-006	Invalid request ID	PATCH with bad id	400/404 error	Medium
REQ-007	UI — submit as facilitator	Facilitator submits new student request	Appears in admin pending list	High

ID	Test Case	Steps	Expected Result	Priority
REQ-008	UI — approve from admin	Super Admin approves request	Student created or request marked approved	High
REQ-009	UI — reject from admin	Admin rejects with reason	Request marked rejected	High
REQ-010	Missing studentData	POST without studentData	400 error	High

13. QR Code

ID	Test Case	Steps	Expected Result	Priority
QR-001	QR generated on student create	Create student with QR_SECRET configured	<code>qr_code</code> field populated	High
QR-002	QR without QR_SECRET	Create student without secret	Student created, QR may be absent	Medium
QR-003	Verify valid QR	POST <code>/api/qr/verify</code> with signed text	<code>{ uniqueId }</code> returned	High
QR-004	Verify invalid QR	POST with tampered text	400 Invalid QR code	High
QR-005	Verify empty text	POST with empty <code>text</code>	400 text is required	Medium
QR-006	UI — display QR on profile	View student with QR	QR image renders, scannable	High

14. Reports & Export

ID	Test Case	Steps	Expected Result	Priority
RPT-001	Admin reports page	Super Admin → Reports	Page loads with filters	Medium
RPT-002	HR reports page	HR Admin → Reports	Attendance stats visible	Medium
RPT-003	Education reports	Education Admin → Reports	Education data visible	Medium
RPT-004	Facilitator attendance reports	Attendance Facilitator → Reports	Grade-scoped data	Medium
RPT-005	Export Excel — attendance	Click export on attendance report	Valid .xlsx file	Medium

ID	Test Case	Steps	Expected Result	Priority
RPT-006	Export PDF — attendance	Click PDF export	Valid PDF file	Medium
RPT-007	Filter reports by date range	Set start/end Ethiopian dates	Data filtered correctly	Medium
RPT-008	Filter reports by grade	Select specific grade	Only that grade's data	Medium

15. Negative & Edge Cases

ID	Test Case	Steps	Expected Result	Priority
NEG-001	Invalid MongoDB ObjectId	Use malformed ID in URL	400 Invalid ID	Medium
NEG-002	Non-existent resource	GET student/result with valid but missing ID	404 Not Found	Medium
NEG-003	SQL/NoSQL injection in email	Login with ' OR 1=1 --	Login fails safely	High
NEG-004	XSS in student name	Create student with <script> in name	Stored safely, rendered escaped in UI	High
NEG-005	Large payload — photo	Upload very large base64 photo	Graceful error or size limit	Low
NEG-006	Concurrent attendance submit	Two facilitators submit same date simultaneously	No duplicate rows (distributed lock)	Medium
NEG-007	Concurrent student create	Two creates with same Unique_ID	Only one succeeds	Medium
NEG-008	API without Content-Type	POST JSON without header	Handled gracefully	Low

16. Test Execution Checklist

Pre-requisites

- ☐ MongoDB running and seeded (or empty for setup flow)
- ☐ Environment variables set (MONGODB_URI, NEXTAUTH_SECRET, optional QR_SECRET)
- ☐ Application running (npm run dev or deployed staging URL)
- ☐ Postman collection & environment imported
- ☐ Test accounts created for all roles

Smoke Test (15 min)

- ☐ AUTH-001 — Login as Super Admin

2. ☐ STU-006 — Create a test student
3. ☐ ATT-004 — Mark attendance for that student
4. ☐ ATT-003 — Verify attendance summary
5. ☐ FAC-003 — Create a test facilitator
6. ☐ AUTH-008 — Logout

Regression Test (Full)

Run all **High** priority cases across modules. Record pass/fail in the test log below.

17. Test Log Template

Date	Tester	Build/Commit	Module	Case ID	Result (Pass/Fail/Blocked)	Notes / Bug ID
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18. Defect Severity Guidelines

Severity	Definition	Example
Critical	System unusable, data loss, security breach	Cannot login; attendance data deleted
High	Major feature broken, no workaround	Cannot create students; wrong RBAC access
Medium	Feature partially broken, workaround exists	Export missing one column; UI layout issue
Low	Cosmetic or minor inconvenience	Typo in label; slow load on one page

19. Postman Collection Usage

1. Import `Birhane-Hiwot-API.postman_collection.json` and `Birhane-Hiwot.postman_environment.json`.
2. Set `baseUrl` and credentials in the environment.
3. Run **Auth** → **Get CSRF Token**, then **Auth** → **Login (Credentials)**.
4. Execute module folders in order: Setup → Students → Attendance → etc.
5. Collection scripts auto-save `studentId`, `facilitatorId`, `teacherId`, `resultId` to environment variables after create requests.

Suggested Newman CLI Run

```
newman run docs/qa/Birhane-Hiwot-API.postman_collection.json \
  -e docs/qa/Birhane-Hiwot.postman_environment.json \
  --folder "Auth" \
  --folder "Students" \
  --folder "Attendance"
```

20. Appendix — API Endpoint Reference

Method	Endpoint	Auth / Role Notes
GET/POST	/api/auth/*	NextAuth handlers
POST	/api/setup/super-admin	One-time setup
GET/POST/DELETE	/api/students	Role checks on POST
GET/PUT/PATCH	/api/students/[id]	—
POST	/api/students/check-duplicate	—
POST	/api/students/count	—
GET	/api/students/total	—
GET	/api/students/[id]/qr	—
GET	/api/students/[id]/payments	—
GET/POST	/api/attendance	—
GET	/api/attendance/[studentId]	—
GET	/api/attendance/temp	—
GET/POST/PUT/DELETE	/api/facilitators	JWT role required
PATCH	/api/facilitators/[id]	canAddStudent toggle
GET	/api/facilitators/total	—
GET/POST/PUT/DELETE	/api/education-facilitators	Education Admin / Super Admin
GET/POST/PUT/DELETE	/api/teachers	Education Admin / Super Admin
GET/POST	/api/subjects	—
POST	/api/subjects/batch	—
DELETE	/api/subjects/[id]	—
GET/POST	/api/teacher-assignments	—
DELETE	/api/teacher-assignments/[id]	—
GET/POST	/api/student-results	—
GET/PUT/DELETE	/api/student-results/[id]	—
GET/POST/PATCH	/api/student-requests	—
GET	/api/student-requests/[id]	—
GET/POST	/api/payment	—
GET/POST/PUT/DELETE	/api/admin-users	Super Admin only
POST	/api/qr/verify	—
GET	/api/sheet/health	—

Method	Endpoint	Auth / Role Notes
GET	/api/sheet	—
GET	/api/cron/aggregate-attendance	Cron job

Document maintained by QA team. Update when new features or API routes are added.